32.4: Views of animal phylogeny continue to be shaped by new molecular and morphological data

As animals with diverse body plans radiated during the early Cambrian, some lineages arose, thrived for a period of time, and then became extinct, leaving no descendants. However, by 500 million years ago, most animal phyla with members alive today were established. Next, we'll examine relationships among these taxa along with some remaining questions that are currently being addressed using genomic data.

The Diversification of Animals

Zoologists currently recognize about three dozen phyla of extant animals, 15 of which are shown in **Figure 32.11**. Researchers infer evolutionary relationships among these phyla by analyzing whole genomes, as well as morphological traits, ribosomal RNA (rRNA) genes, *Hox* genes, protein-coding nuclear genes, and mitochondrial genes. Notice how the following points are reflected in Figure 32.11.

Figure 32.11

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A phylogeny of living animals.
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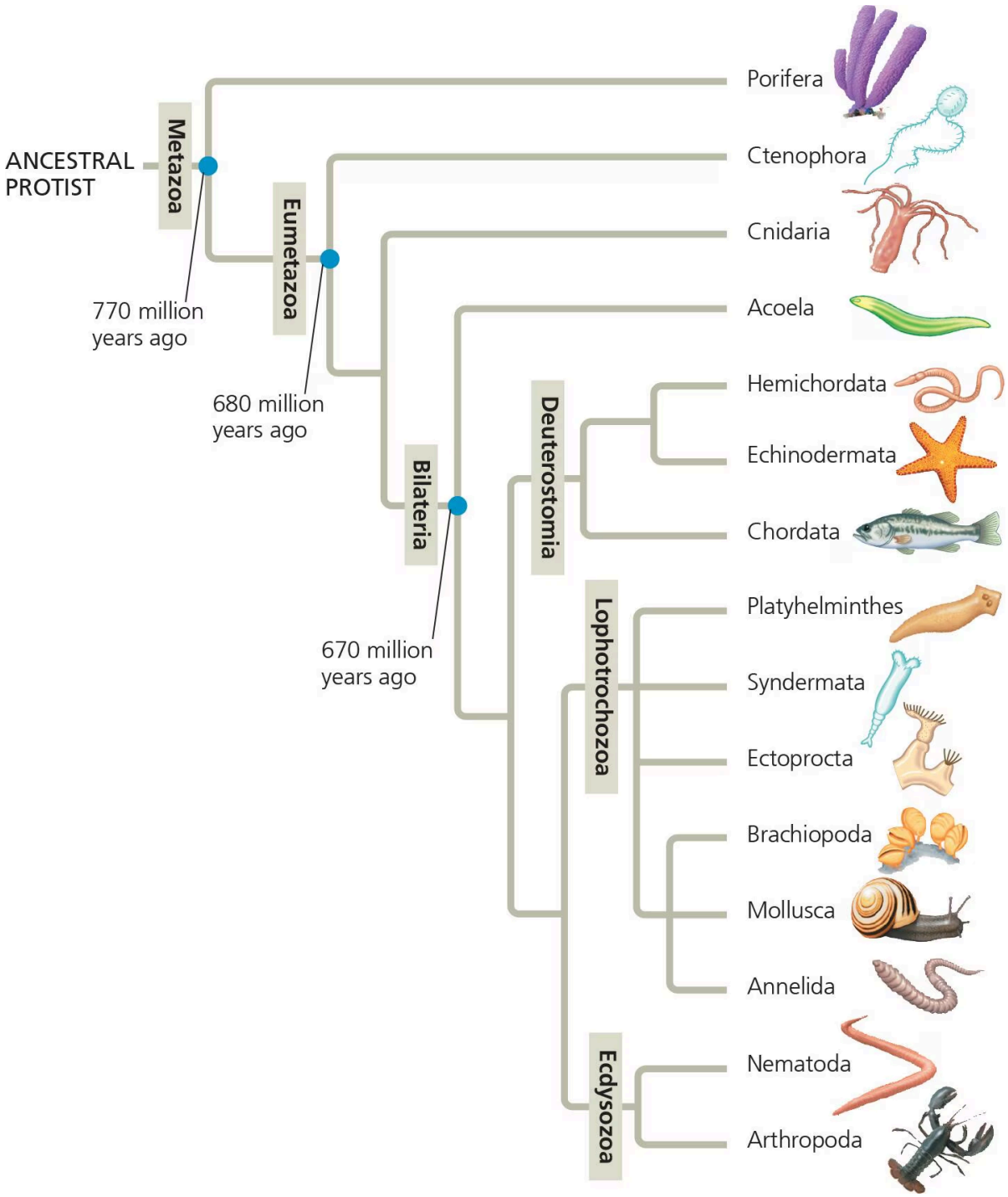

This phylogeny shows a leading hypothesis about the relationships among selected animal phyla. The bilaterians are divided into three main lineages: deuterostomes, lophotrochozoans, and ecdysozoans. The dates of origin identified here are based on the results of a recent molecular clock study.

Phylogeny of living animals. Tree is described as a list. Ancestral Protist (Metazoa):

Branch Point 1 (770 million years ago): Porifera.

Branch Point 2 (Eumetazoa) (680 million years ago): Ctenophora.

Branch Point 3: Cnidaria.

Branch Point 4 (Bilateria) (670 million years ago): Acoela.

Branch Point 5.

Branch Point 6 (Deuterostomia).

Branch Point 7. Hemichordata. Echinodermata. Chordata.

Branch Point 8.

Branch Point 9 (Lophotrochozoa). Platyhelminthes. Syndermata. Ectoprocta.

Branch point 10: Brachiopoda. Mollusca. Annelida.

Branch Point 11 (Ecdysozoa). Nematoda. Arthropoda.

VISUAL SKILLS*Which phylum is the sister group of Bilateria in this tree?*

*For suggested answer, see*Appendix A*.*

- **All animals share a common ancestor.** Current evidence indicates that animals are monophyletic, forming a clade called Metazoa. All extant and extinct animal lineages have descended from a common ancestor.
- **Sponges are the sister group to all other animals.** Sponges (phylum Porifera) are basal animals, having diverged from all other animals early in the history of the group. Recent morphological and phylogenomic analyses indicate that sponges are monophyletic, as shown here.
- **Eumetazoa is a clade of animals with tissues.** All animals except for sponges and a few others belong to a clade of **eumetazoans** (“true animals”). Members of this group have tissues, such as muscle tissue and nervous tissue. Basal eumetazoans, which include the phyla Ctenophora (comb jellies) and Cnidaria, are diploblastic and generally have radial symmetry.
- **Most animal phyla belong to the clade Bilateria.** Bilateral symmetry and the presence of three prominent germ layers are shared derived characters that help define the clade Bilateria. This clade contains the majority of animal phyla, and its members are known as *bilaterians*. The Cambrian explosion was primarily a rapid diversification of bilaterians.
- **There are three major clades of bilaterian animals.** Bilaterians have diversified into three main lineages, Deuterostomia, Lophotrochozoa, and Ecdysozoa. With one exception, the phyla in these clades consist entirely of **invertebrates**, animals that lack a backbone; Chordata is the only phylum that also includes **vertebrates**, animals with a backbone.

As depicted in Figure 32.11, hemichordates (acorn worms), echinoderms (sea stars and relatives), and chordates are members of the bilaterian clade **Deuterostomia**; thus, the term *deuterostome* refers not only to a mode of animal development, but also to the members of this clade. (The dual meaning of this term can be confusing since some organisms with a deuterostome developmental pattern are *not* members of clade Deuterostomia.) Hemichordates share some characteristics with chordates, such as gill slits and a dorsal nerve cord; echinoderms lack these characteristics. These shared traits may have been present in the common ancestor of the deuterostome clade (and lost in the echinoderm lineage). As mentioned above, phylum Chordata, the only phylum with vertebrate members, also includes invertebrates.

Bilaterians also diversified in two major clades that are composed entirely of invertebrates: the *lophotrochozoans* and the *ecdysozoans*. The name **Lophotrochozoa** refers to two different features observed in some animals belonging to this clade **(Figure 32.12)**. Some lophotrochozoans, such as ectoprocts, develop a unique structure called a **lophophore** (from the Greek *lophos*, crest, and *pherein*, to carry), a crown of ciliated tentacles that function in feeding (see Figure 32.12a). Individuals in other phyla, including molluscs and annelids, go through a distinctive developmental stage called the **trochophore larva** (see Figure 32.12b)—hence the name lophotrochozoan.

Figure 32.12

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Morphological characteristics of lophotrochozoans.
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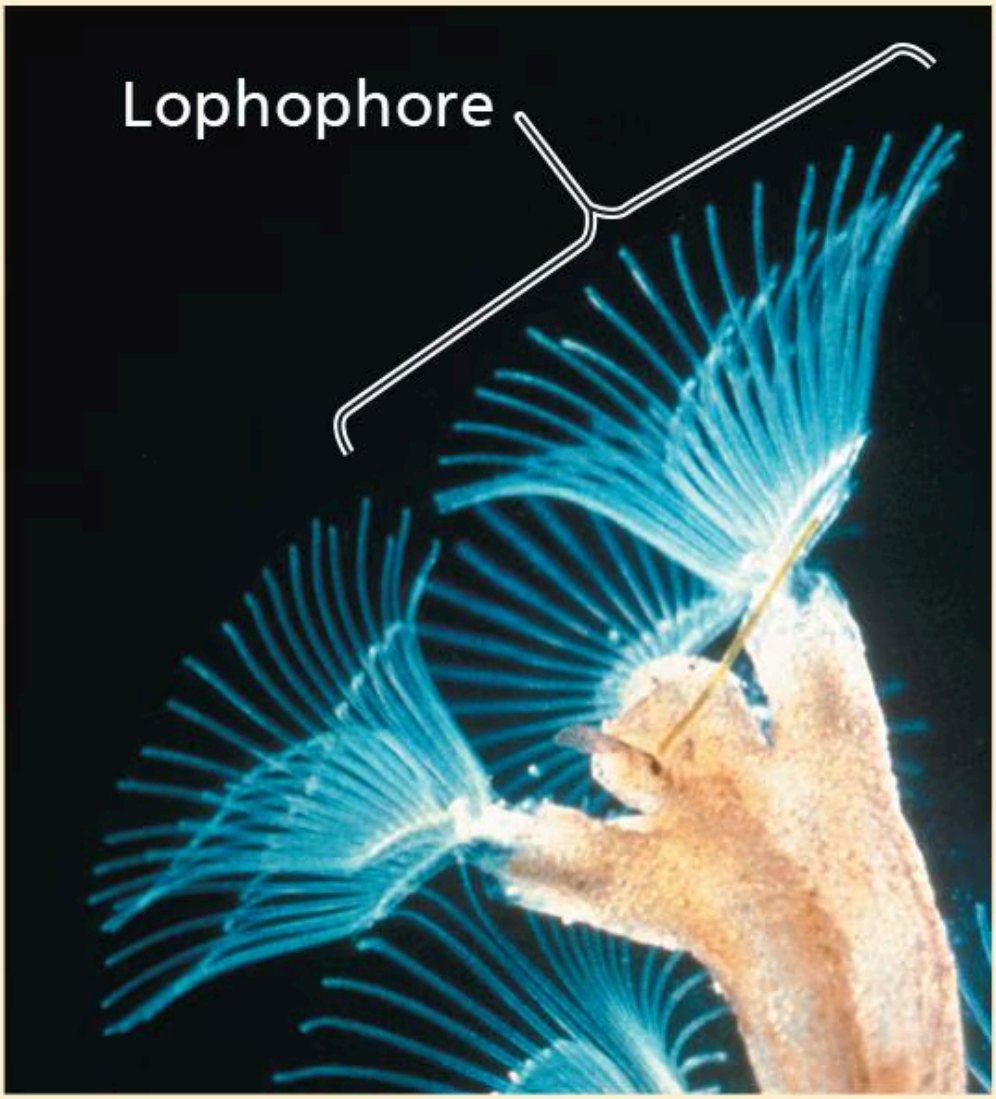
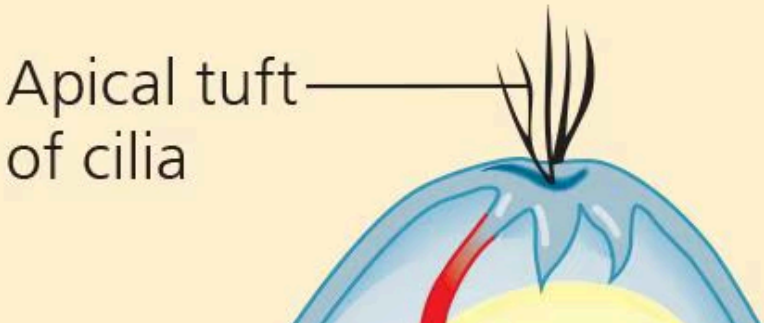
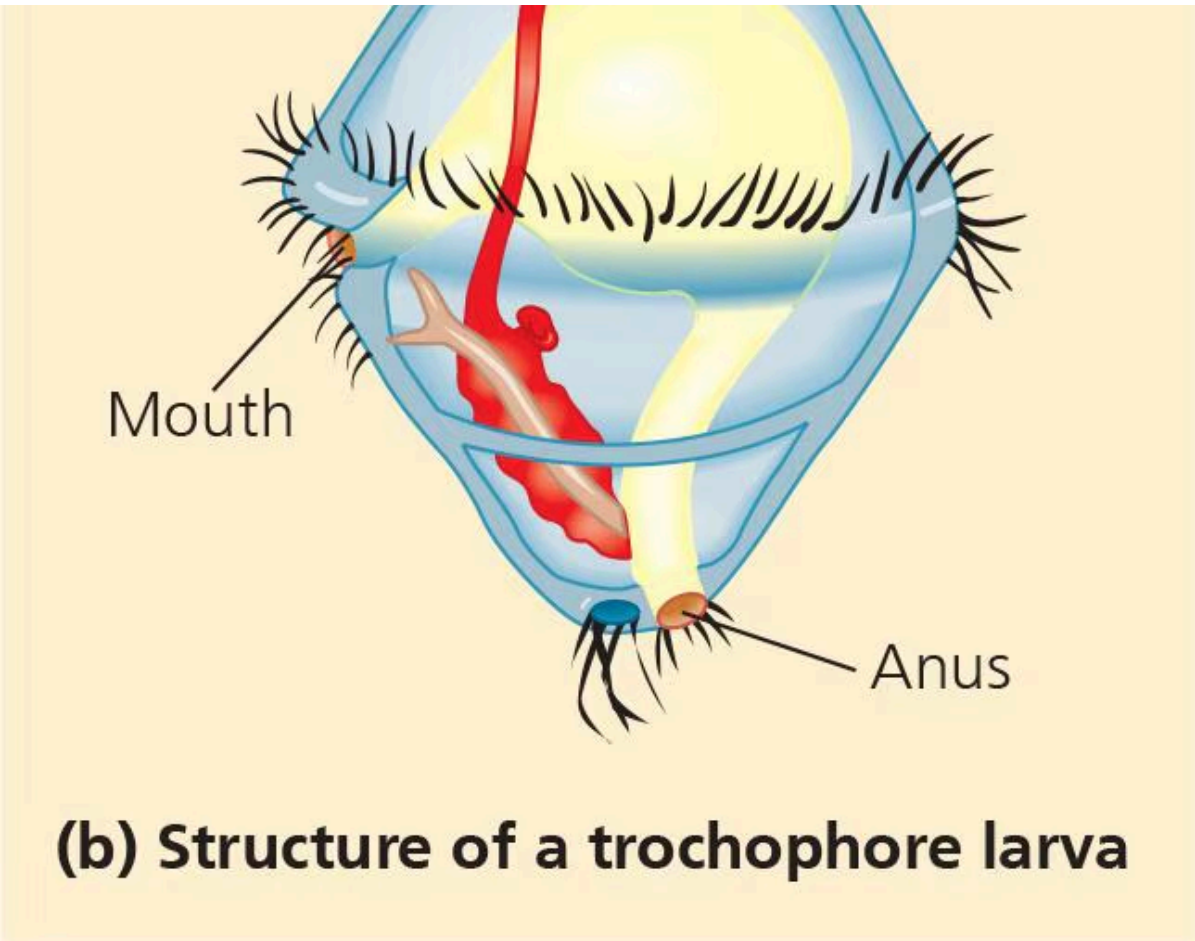


(a) Lophophore feeding structures of an ectoproct





Part a. Lophophore feeding structures of an ectoproct showing its crown of ciliated tentacles that function in feeding.

Part b. Structure of a trochophore larva which has a teardrop shape, with an apical tuft of cilia at the top, the mouth on the side and an anus at the bottom.

The clade name **Ecdysozoa** refers to a characteristic shared by nematodes, arthropods, and some of the other ecdysozoan phyla that are not included in our survey. These animals secrete external skeletons (exoskeletons); the stiff covering of a cricket and the flexible cuticle of a nematode are examples. As the animal grows, it molts, squirming out of its old exoskeleton and secreting a larger one. The process of shedding the old exoskeleton is called *ecdysis*. Though named for this characteristic, the clade was proposed mainly on the basis of molecular data that support the common ancestry of its members. Furthermore, some taxa excluded from this clade by their molecular data, such as certain species of leeches, do in fact molt.

Future Directions in Animal Systematics

While many scientists think that current evidence supports the phylogeny in Figure 32.11, aspects continue to be debated. Although it can be frustrating that the phylogenies in textbooks are not set-in-stone truths, the uncertainty inherent in these diagrams is a healthy reminder that science is an ongoing, dynamic process

of inquiry. We'll conclude with two questions that are the focus of ongoing research.

- **Are ctenophores basal metazoans?** Many researchers have concluded that sponges are basal metazoans (see Figure 32.11). This conclusion was supported in a 2017 phylogenomic analysis, but several other studies have placed the comb jellies (phylum Ctenophora) at the base of the animal tree. In addition to the most recent phylogenomic results, data consistent with placing sponges at the base of the animal tree include fossil steroid evidence, molecular clock analyses, the morphological similarity of sponge collar cells to the cells of choanoflagellates (see Figure 32.3), and the fact that sponges are one of the few animal groups that lack tissues (as might be expected for basal animals). Ctenophores, on the other hand, have tissues, and their cells do not resemble the cells of choanoflagellates. At present, the idea that ctenophores are basal metazoans remains an intriguing but controversial hypothesis.
- **Are the Acoela basal bilaterians?** A series of recent papers—including a 2016 phylogenomic study—have indicated that flatworms in phylum Acoela are basal bilaterians, as shown in Figure 32.11. A different conclusion was supported by one recent analysis, which placed members of the Acoela within Deuterostomia. Researchers are sequencing the genomes of additional species in the phylum and in closely related groups to provide a more definitive test of the hypothesis that the Acoela are basal bilaterians. If further evidence supports this hypothesis, bilaterians may have descended from a common ancestor that resembled this group of living flatworms—that is, an ancestor that had a simple nervous system, a saclike gut with a single opening (the "mouth"), and no excretory system.